Calculus II HW 3

Part I. Regular problems

1. Determine the radius and interval of convergence of the following series:

(a) $\sum_{k=1}^{\infty} \frac{(-1)^k}{k \cdot 3^k}(x-1)^k$

(b) $\sum_{k=1}^{\infty} \frac{(k!)^2}{(2k)!}x^{2k+1}$

(c) $\sum_{k=0}^{\infty} k!(x+1)^k$

(d) $\sum_{k=1}^{\infty} \frac{(-1)^k}{\sqrt{k}}(3x-1)^k$

(e) $\sum_{k=2}^{\infty} (k+3)^2(2x-3)^k$

2. Find a power series representation of $f(x)$ about $c = 0$. Also, determine the radius and interval of convergence.

(a) $f(x) = \frac{2}{1-x}$

(b) $f(x) = \frac{2}{1-x^2}$

(c) $f(x) = \frac{3x}{1+x^2}$

3. Find a power series representation and radius of convergence by integrating or differentiating one of the series from Q2.

(a) $f(x) = 2\ln(1-x)$

(b) $f(x) = \dfrac{2x}{(1-x^2)^2}$

(c) $f(x) = \ln(1+x^2)$

Part II. Group problems

1. Find the radius of convergence and interval of convergence of the power series.
   (a) $\sum_{n=1}^{\infty} (-1)^n n x^n$
   (b) $\sum_{n=2}^{\infty} \dfrac{5^n}{n} x^n$
   (c) $\sum_{n=1}^{\infty} \dfrac{x^{2n}}{n!}$

2. (a) Find a power series representation of $f(x)$ about $c = 0$. Also, determine the radius and interval of convergence.

(i) $f(x) = \dfrac{3}{1+x^2}$

(ii) $f(x) = \dfrac{3}{x-1}$

(b) Find a power series representation and radius of convergence by integrating or differentiating one of the series from (a).

(i) $f(x) = 3\tan^{-1} x$

(ii) $f(x) = \dfrac{3}{(x-1)^2}$